Skills Summary

Multi-Theorem Circle Configurations

Use this reference while completing your worksheet or when studying.

Skill 1 — Inscribed and central angles

Inscribed angle: an inscribed angle is *half* the central angle standing on the same arc: $m\angle ACB = \frac{1}{2}\, m\angle AOB$.

Angle in a semicircle: an angle inscribed on a diameter is a right angle.

Cyclic quadrilateral: opposite angles are supplementary (they add to 180°).

Arc measure equals the measure of its central angle.

Example: In circle O, central $\angle AOB = 100^\circ$ and C lies on the major arc. Find $m\angle ACB$.
Step 1. The angle I want is inscribed and the angle I know is central, and both stand on the same arc AB — that is exactly the inscribed-angle theorem.
Step 2. Inscribed = half of central: $m\angle ACB = \frac{1}{2}(100^\circ)$. $\Rightarrow$ **50** degrees

Try it: In circle O, central $\angle AOB = 130^\circ$ and C lies on the major arc. Find $m\angle ACB$.

check: **65** degrees

Skill 2 — Arc length and sector area

A central angle of θ degrees cuts off $\dfrac{\theta}{360}$ of the circle, so it takes that same fraction of the circumference and of the area:

Arc length $= \dfrac{\theta}{360} \cdot 2\pi r$ **Sector area** $= \dfrac{\theta}{360} \cdot \pi r^2$

Segment area = sector area − area of the triangle formed by the two radii and the chord.

Example: A circle has radius 6 cm and a central angle of 120°. Find the arc length to the nearest tenth of a centimetre.
Step 1. I want a length along the curve, not an area, so I take the angle's fraction of the circumference.
Step 2. $\dfrac{120}{360} \cdot 2\pi(6) = \frac{1}{3}(12\pi) = 4\pi \approx 12.566$ cm. $\Rightarrow$ **12.6** cm

Try it: A circle has radius 9 cm and a central angle of 60°. Find the arc length to the nearest tenth of a centimetre.

check: **9.4** cm

Skill 3 — Chords, tangents and secants

Two chords meeting inside at P: $AP \cdot PB = CP \cdot PD$ — the two products of the pieces are equal.

Tangent and secant from an outside point P: $PT^2 = PA \cdot PB$, where PB is the *whole* secant, not just the far piece.

Two secants from an outside point: $PA \cdot PB = PC \cdot PD$.

Example: Chords AB and CD meet inside a circle at P with $AP = 4$, $PB = 9$, $CP = 3$ and $PD = x$. Find x.
Step 1. The intersection point is inside the circle and I know three of the four pieces, so the intersecting-chords theorem gives an equation in x.
Step 2. $3x = 4 \cdot 9 = 36$, so $x = 36 \div 3$. $\Rightarrow$ $\boldsymbol{x = 12}$

Try it: Chords AB and CD meet inside a circle at P with $AP = 5$, $PB = 8$, $CP = 4$ and $PD = x$. Find x.

check: $\boldsymbol{x = 10}$

Skill 4 — The equation of a circle

Centre-radius form: $(x-h)^2 + (y-k)^2 = r^2$ has centre (h, k) and radius r — read the centre with the signs *flipped*.

To complete the square: halve the coefficient of x, square it, and add it to both sides; do the same for y. Then the radius is the square root of the right-hand side.

Example: Find the centre and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$.
Step 1. The equation has x and y terms but no squared binomials, so I complete the square in each variable to reach centre-radius form.
Step 2. Half of 4 is 2 and half of 6 is 3: $(x-2)^2 + (y+3)^2 = 12 + 4 + 9 = 25$, and $r = \sqrt{25}$.
$\Rightarrow$ centre $\mathbf{(2, -3)}$, radius $\boldsymbol{r = 5}$

Try it: Find the centre and radius of $x^2 + y^2 - 8x + 2y - 8 = 0$.

check: centre $\mathbf{(4, -1)}$, radius $\boldsymbol{r = 5}$